

The dual simplex method on a tableau

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In this note we demonstrate by example how the dual simplex method can be implemented by performing row operations on the simplex tableau. Consider the following linear optimization problem (P):

$$\begin{aligned} &\text{Minimize } Z_P = 7x_1 + 2x_2 + 5x_3 + 4x_4 \\ &\text{Subject to} \\ &\quad 2x_1 + 4x_2 + 7x_3 + x_4 - x_5 = 5 \quad (y_1) \\ &\quad 8x_1 + 4x_2 + 6x_3 + 4x_4 - x_6 = 8 \quad (y_2) \quad (P) \\ &\quad 3x_1 + 8x_2 + x_3 + 4x_4 - x_7 = 4 \quad (y_3) \\ &\quad x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0, \end{aligned}$$

and the corresponding dual (D) of (P) given by:

$$\begin{aligned} &\text{Maximize } Z_D = 5y_1 + 8y_2 + 4y_3 \\ &\text{Subject to} \\ &\quad 2y_1 + 8y_2 + 3y_3 \leq 7 \quad (x_1) \\ &\quad 4y_1 + 4y_2 + 8y_3 \leq 2 \quad (x_2) \\ &\quad 7y_1 + 6y_2 + y_3 \leq 5 \quad (x_3) \\ &\quad y_1 + 4y_2 + 4y_3 \leq 4 \quad (x_4) \quad (D) \\ &\quad -y_1 \leq 0 \quad (x_5) \\ &\quad -y_2 \leq 0 \quad (x_6) \\ &\quad -y_3 \leq 0 \quad (x_7) \end{aligned}$$

A basis is a subset $B \subseteq \{1, \dots, 7\}$ of three x -variables. The primal basic solution $\bar{x}$ associated with B is obtained by first fixing the non-basic x -variables to zero, and then solving the equalities for the basic x -variables.

The dual basic solution $\bar{y} \in \mathbb{R}^3$ associated with B is obtained by setting the inequalities of (D) corresponding to basic x -variables to equality, and then solve the system for the y -variables.

For the basis $B = \{5, 6, 7\}$, the primal basic solution is given by $\bar{x}^T = [0, 0, 0, 0, -5, -8, -4]$, and the dual basic solution is given by $\bar{y}^T = [0, 0, 0]$. Both the basis matrix A_B and basis inverse A_B^{-1} are minus the identity matrix, *i.e.*, one has $A_B = A_B^{-1} = -I_3$. Observe that $\bar{x}$ is not feasible for (P), but that $\bar{y}$ is feasible for (D). Hence one may use the dual simplex method starting from the basis B to solve (P) and (D).

First construct the simplex tableau associated with B . Write the equations $-Z_P + c^T x = 0$ and $Ax = b$ in tableau form:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1	7	2	5	4	0	0	0	0
x_5	0	2	4	7	1	-1	0	0	5
x_6	0	8	4	6	4	0	-1	0	8
x_7	0	3	8	1	4	0	0	-1	4

Now perform row operations on the equalities above to find the tableau corresponding to B , *i.e.*, solve the equalities $-Z_P + c^T x = 0$ and $Ax = b$ wrt. Z_P and the basic variables. To achieve this, multiply the last three rows with (-1) to obtain the table:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1	7	2	5	4	0	0	0	0
x_5	0	-2	-4	-7	-1	1	0	0	-5
x_6	0	-8	-4	-6	-4	0	1	0	-8
x_7	0	-3	-8	-1	-4	0	0	1	-4

Observe that columns corresponding to basic variables are unit vectors. Also observe that the values of the dual slack variables (the reduced costs) $\bar{s} := c - A^T \bar{y}$ appear in the first row of the simplex tableau. For the basis $B = \{5, 6, 7\}$, the vector $\bar{s}$ is therefore given by $\bar{s}^T = [7, 2, 5, 4, 0, 0, 0]$. Since $\bar{s} \geq 0_7$, the dual basic solution $\bar{y}$ is feasible for (D). Also, since the basic solution $\bar{x}$ is *not* non-negative, the basis B is not feasible for (P).

In the dual simplex method, one first selects a basic variable from the basic variables $B = \{i_1, \dots, i_m\}$ to exit the basis, and one must choose a basic variable that has a negative value in the (current) basic solution $\bar{x}$. In our case $B = \{i_1, i_2, i_3\} = \{5, 6, 7\}$, and the values of the basic variables are $\bar{x}_{i_1} = \bar{x}_5 = -5$, $\bar{x}_{i_2} = \bar{x}_6 = -8$ and $\bar{x}_{i_3} = \bar{x}_7 = -4$. We choose the basic variable $x_{i_2} = x_6$ to exit the basis.

Given a chosen $i_k \in \{i_1, \dots, i_m\}$ as the basic variable to exit the basis, the dual simplex method tries to move as far as possible from $\bar{y}$ in the direction $-\bar{b}_k$ while preserving dual feasibility, where $\bar{b}_k$ denotes the k^{th} row of the basis inverse A_B^{-1} . In other words the dual simplex method computes:

$$\begin{aligned}\theta^* &:= \sup\{\theta : \bar{y} - \theta \cdot \bar{b}_k \text{ is feasible for (D)}\} \\ &= \sup\{\theta : A^T(\bar{y} - \theta \cdot \bar{b}_k) \leq c\} \\ &= \sup\{\theta : \bar{s} + \theta \cdot \bar{a}_k \geq 0\},\end{aligned}$$

where $\bar{a}_k = A^T \bar{b}_k$ denotes the k^{th} row of the (updated) matrix $\bar{A} := A_B^{-1} A$.

In our example we have:

$$A = \begin{pmatrix} 2 & 4 & 7 & 1 & -1 & 0 & 0 \\ 8 & 4 & 6 & 4 & 0 & -1 & 0 \\ 3 & 8 & 1 & 4 & 0 & 0 & -1 \end{pmatrix} \quad \text{and} \quad A_B = A_B^{-1} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

and therefore:

$$\bar{A} = A_B^{-1} A = \begin{pmatrix} -2 & -4 & -7 & -1 & 1 & 0 & 0 \\ -8 & -4 & -6 & -4 & 0 & 1 & 0 \\ -3 & -8 & -1 & -4 & 0 & 0 & 1 \end{pmatrix},$$

and therefore, since $i_k = i_2$ in our case, the second row of $\bar{A}$ is $\bar{a}_2^T = [-8, -4, -6, -4, 0, 1, 0]$. The stepsize θ^* can now be computed:

$$\theta^* = \sup\{\theta : \bar{s} + \theta \bar{a}_2 = \begin{pmatrix} 7 \\ 2 \\ 5 \\ 4 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \theta \begin{pmatrix} -8 \\ -4 \\ -6 \\ -4 \\ 0 \\ 1 \\ 0 \end{pmatrix} \geq 0\} = \frac{1}{2}.$$

Which dual constraint prohibits a larger value of θ ? The answer is the second constraint: If we choose $\theta > \frac{1}{2}$, then $\bar{s}_2$ becomes negative. Hence $\bar{y} - \frac{1}{2} \cdot \bar{b}_2$ satisfies the second constraint of $A^T y \leq c$ with equality. Since $\bar{y} - \frac{1}{2} \cdot \bar{b}_2$ also satisfies the inequalities corresponding to the basic variables in $B \setminus \{i_2\} = \{5, 7\}$ with equality, the point $\bar{y} - \frac{1}{2} \cdot \bar{b}_2$ is the basic feasible solution corresponding to the basis:

$$B' := (B \cup \{2\}) \setminus \{6\}.$$

How can we understand this change of the basis in terms of the current simplex tableau?

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1	7	2	5	4	0	0	0	0
x_5	0	-2	-4	-7	-1	1	0	0	-5
x_6	0	-8	-4	-6	-4	0	1	0	-8
x_7	0	-3	-8	-1	-4	0	0	1	-4

Note that all the information we needed (the number $\bar{x}_6$ and the vectors $\bar{s}$ and $\bar{a}_2$) is given in the simplex tableau. Specifically, the parameter $\bar{x}_6$ and the vectors $\bar{s}$ and $\bar{a}_2$ are located in the simplex tableau as follows

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1				$\bar{s}$				0
x_5	0	-2	-4	-7	-1	1	0	0	-5
x_6	0				$\bar{a}_2$				$\bar{x}_6$
x_7	0	-3	-8	-1	-4	0	0	1	-4

Therefore, the question of finding the largest $\theta \geq 0$ such that $\bar{s} + \theta \bar{a}_2 \geq 0_7$ is the same as finding the largest $\theta \geq 0$ one can multiply to the third row and add to the first row, while still maintaining that the first row is non-negative. In addition, one can decide which variable to enter the basis by simply looking at the reduced cost(s) that become zero.

To obtain the (new) simplex tableau corresponding to B' , we perform Gaussian elimination on the tableau corresponding to B so as to make x_2 the basic variable in the row of the tableau corresponding to x_6 . This means making the column of the tableau corresponding to x_2 a unit vector. First multiply the row of the tableau corresponding to x_6 with (-2) , and add the result to the last row of the tableau:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1	7	2	5	4	0	0	0	0
x_5	0	-2	-4	-7	-1	1	0	0	-5
x_6	0	-8	-4	-6	-4	0	1	0	-8
x_7	0	13	0	11	4	0	-2	1	12

Next multiply the row of the tableau corresponding to x_6 with (-1) , and add the result to the row corresponding to x_5 :

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1	7	2	5	4	0	0	0	0
x_5	0	6	0	-1	3	1	-1	0	3
x_6	0	-8	-4	-6	-4	0	1	0	-8
x_7	0	13	0	11	4	0	-2	1	12

Now multiply the row of the tableau corresponding to x_6 with $(\frac{1}{2})$ and add the result to the row of the tableau corresponding to Z_P :

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1	3	0	2	2	0	$\frac{1}{2}$	0	-4
x_5	0	6	0	-1	3	1	-1	0	3
x_6	0	-8	-4	-6	-4	0	1	0	-8
x_7	0	13	0	11	4	0	-2	1	12

The final step is to multiply the row of the tableau corresponding to x_6 with $-\frac{1}{4}$ and declare x_2 to be the basic variable in the row of the tableau that previously had x_6 as the basic variable:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right hand side
Z_P	-1	3	0	2	2	0	$\frac{1}{2}$	0	-4
x_5	0	6	0	-1	3	1	-1	0	3
x_2	0	2	1	$\frac{3}{2}$	1	0	$-\frac{1}{4}$	0	2
x_7	0	13	0	11	4	0	-2	1	12

This is now the simplex tableau associated with $B' = \{5, 2, 7\}$. We see that the primal basic solution associated with B' is $\bar{x}' = [0, 2, 0, 0, 3, 0, 12]$ (from the right-hand-sides in the tableau). Since $\bar{x}' \geq 0_7$, the basis B' is primal feasible. Furthermore, the first row of the tableau gives that $\bar{s}' = [3, 0, 2, 2, 0, \frac{1}{2}, 0]$ is the vector of values of the dual slack variables associated with B' . Hence the dual basic solution $\bar{y}'$ associated with B' is also feasible for (D). From the Strong Duality Theorem one obtains that B' is an optimal basis for (P) and (D).

To obtain the dual basic solution $\bar{y}'$ corresponding to B' , set the dual slack variables corresponding to the basic variables x_2, x_5 and x_7 to zero, and solve for $\bar{y}'$. Specifically one solves the equations:

$$\begin{aligned} 4y_1 + 4y_2 + 8y_3 &= 2 & (x_2) \\ -y_1 &= 0 & (x_5) \\ -y_3 &= 0 & (x_7) \end{aligned}$$

which gives the optimal dual basic solution $\bar{y}' = [0, \frac{1}{2}, 0]$.